

A foundation model for atomistic materials chemistry: MACE-MP-0

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Abstract

Machine Learning Potentials (MLPs) have revolutionized the field of material science. They provide a computational framework to simulate the atomistic motion of systems with the computational efficiency of traditional force fields combined with the quantum-level accuracy and reactivity of *ab initio* techniques. This makes MLPs a powerful tool for investigating processes that are challenging to conduct *in vitro* and that require a high level of resolution of insight. However, state-of-the-art MLPs require significant human and computational resources to develop and validate a model for a particular system of interest. The MACE-MP-0 is a remarkable pre-trained foundational model designed to simulate a wide range of atomic systems without needing system-specific retraining. This report investigates the applicability and the limitations of MACE-MP-0 by simulating protein dynamics in vacuum and solvated environments. This work found that MACE-MP-0 is able to simulate a mini-protein’s structural stability and early folding dynamics, capturing reactive processes such as protonations. However, it was also found that it is challenging for the model to simulate protein dynamics in bulk-water systems due to its computational limitations.

1. Introduction

1.1. Overview of Atomistic Simulation Techniques

Predicting the properties of materials at different lengths and timescales, such as thermodynamic stability, mechanical features, phase transition or nanoscale structure, is a crucial yet challenging task in the field of material science. [1] Atomistic simulation techniques revolutionized and accelerated the field of material discovery, design and optimization by providing atomistic insights into chemical processes. The first breakthrough was the introduction of Molecular Dynamics by Alder in 1957 [2] and Wainwright in 1964 [3], which allows for the simulation of the motion of a system. This is achieved by discretizing and numerically calculating Newton’s equations of motion:

$$m_i \frac{d^2 \mathbf{r}_i}{dt^2} = -\nabla_i U \quad (1)$$

where m_i is the mass of particle i the system, r_i is its position, and U is the potential energy of the system. The positions of the molecules at each timestep dt are updated, producing a trajectory of the system. The value of the timestep of the simulation should be chosen carefully so it is small enough that it can trace the fastest dynamics of the system, while the duration of the simulation has to be large enough to capture the slowest

dynamics. Calculating the system’s trajectory in a statistical ensemble allows for the microscopic sampling of its phase space from which the macroscopic thermodynamic properties can be obtained using the laws of statistical mechanics. [4] Molecular dynamics allows for conducting transformative experiments *in silico* that are difficult to achieve *in vitro*. For example, molecular dynamics have brought a breakthrough in the research of protein-ligand binding mechanism [5], membrane proteins [6] or the flow of nano-fluids [7].

One of the key challenges of developing a molecular dynamics simulation is building a model that accurately describes the interparticle interactions governed by the potential energy surface of the system (PES), $U(r_1, r_2, \dots, r_n)$. A highly accurate method for such calculation is the *ab initio* method of Density Functional Theory (DFT) [8]. DFT calculates the electronic structure of the system at each timestep from first principles by approximating the Schrödinger equation. While this method is highly accurate, it is computationally prohibitive at larger systems or timescales due to the many-body repulsive interactions of the electrons. [9] A solution for this scalability problem is the use of classical force fields, for example, popular choices are the AMBER [10] or CHARMM [11] force fields. These classical force fields model the molecular interactions with physically motivated, empirical functions whose param-

eters are fitted to in-vivo or in-vitro experiments. For example, covalent bonds are often described by harmonic spring forces, or the long-range electrostatic interactions with the Lennard-Jones potential [12]. While force fields have higher computational efficiency than DFT, they lack the accuracy to describe complex systems and their electronic structure. Additionally, force fields assume the bonds to be constant, making the system non-reactive, prohibiting the simulation of processes that require bond breaking, electron excitation or proton transfer. [13]

1.2. Machine Learning Potentials and Their Purpose

In the past decade, the transformative approach of Machine Learning Potentials (MLPs) has revolutionized the field of atomistic simulations. [14] MLPs are supervised machine learning models that can predict the PES of a system from its structure with *ab initio* accuracy and with the computational efficiency of classical force fields. [9] [15] MLPs have enabled computationally fast and accurate simulations across a wide range of lengths and time scales, providing atomistic insights into material properties that would be too expensive or complex to calculate at first principle accuracy.

A key use for MLPs is the simulation of reactive processes. Force fields are inherently non-reactive as they assume the bonds to be constant, while *ab initio* methods are too computationally expensive to perform a converged sampling of reactive processes. Since in the case of MLPs, no topology is defined, MLPs are reactive, therefore they can be used to simulate reactive processes or dissociation. For example, at the atomic scale, MLPs allowed for the simulation of the dissociation of water at catalytic interfaces [16] or the decomposition of molecules at air-water boundaries [17].

MLPs also allowed performing experiments *in silico* that cannot be achieved *in vitro* using current technologies. An example of this is the simulation of nanoconfined water between two carbon monolayers [18], which revealed two new phases of water (hexatic and superionic) that were previously inaccessible to *ab initio* molecular dynamics due to computational limitations. MLPs have also been used to model mesoscale processes, for example, simulating water flow through nanotubes, predicting material-dependent friction effects. [19] At the macroscopic scale, general purpose MLPs have been developed that are able to simulate all phases and structure of one specific element, for example, silicon [20] and carbon [21], reproducing experimental observations of phase transitions and discovering new phases under extreme conditions.

1.3. Key Features of MLPs

There are four key steps in the development of accurate and efficient MLPs: (a) generating high-quality training datasets for training, (b) defining chemical descriptors that contain structural information of the system suitable for supervised learning, (c) choosing a regression model that maps the chemical descriptors to atomic energies such that they sum up to the total energy and (c) methods for validating the resulting PES. During supervised learning, the structural information representative of the described system is assembled, and the ML model learns how to interpolate between points in their neighbourhood. Quantum level accuracy in the prediction can be achieved by training the model on *ab initio* reference data that includes structural and PES information, while the computational efficiency arises from the lower computational cost of the interpolation process compared to DFT calculations. [22]

Chemical descriptors are a key step in constructing MLPs. They build a fingerprint of the chemical neighbourhood around each atom in the system by transforming the cartesian coordinates of atomic structures into a more representative form for supervised learning. Chemical descriptors need to preserve the rotational, translational and permutational invariance for the accurate prediction of the PES, as the energy of a structure has to obey these invariances. Chemical descriptors are usually overcomplete, aiding the regression of ML models by increasing their sensitivity. Descriptions of the atomic environments are usually truncated after a certain cutoff to avoid large computational costs, assuming that the structures outside the cutoff do not have a significant contribution to the total energy. [22]

1.4. Overview of MLPs Models

The first MLP, Neural Network Potentials (NNPs), was introduced by Doren et al. in 1995. [23] NNPs employ a feed-forward neural network for the regression task, predicting the atomic energies of the input molecular structures. Early NNPs have significantly reduced the computational costs of MD simulations, however, they lacked computational accuracy. [24] In 2007, Behler and Parrinello introduced the High-Dimensional Neural Network Potentials (HD-NNPs) that have increased accuracy compared to NNPs. [25] HD-NNPs use a multi-layered, forward-passing neural network for each element in the system to predict atomic energies. HD-NNPs employed the chemical descriptor Atom-Centered Symmetry Functions (ACSFs), developed by Jörg Behler [26].

Another seminal technique that has emerged in the field of MLPs is Kernel-based regression approaches

that use kernel functions for similarity measurements between representative locations. Bartók et al. in 2010 employed this methodology and developed the Gaussian Approximation Potentials (GAP) [27], using the Smooth Overlap of Atomic Positions (SOAP) technique as chemical descriptors for similarity measurements, achieving quantum accuracy in the simulations. [27] [28] A further development in this field was the introduction of Moment Tensor Potentials (MTPs) by Shapeev in 2010 [29]. MTPs use moment tensors as chemical descriptors, which capture the rearrangements of the atoms dynamically, allowing for a systematical improvement of the atomic representations. These techniques preserve the required invariances and have an analytical, smooth form that enables calculating derivatives. However, the descriptors of these models have inherent limitations, such as they scale with the number of atomic species and they are incomplete due to only having two-body terms. Atomic Cluster Expansion (ACE) [30] overcome these limitations by generating higher body terms at a constant computational cost per basis function, making them complete and linear-scalable descriptors. ACE also ensures that the vectorial and tensorial output properties of the system have equivariance, for example, the dipole moment and polarizability of the system transform accurately if the coordinates are rotated.

1.5. Limitations of MLPs

An inherent limitation of all models described so far is that they are short-sighted due to the cutoff applied around each atom. This locality approximation usually holds well as chemical bonds are mainly determined by the local environment, however, in systems where long-range interactions are significant, short-sightedness can cause poor quality predictions. This limitation can be overcome by learning partial charges during the training process [31] [32] or using long-range descriptors (LODE) [35]. However, modern approaches employ descriptors that develop the representation of the atomic environment during the training process using message-passing neural networks (MP-NNs), such as SchNet [33], PhysNet [34], NequiP [35], Allegro [36] or (MACE) [37], which enables capturing both local and long-range atomic interactions efficiently.

2. Methodology

2.1. Equivariant Message Passing Neural Network Potentials (MP-NNPs)

State-of-the-art models use Message-Passing Neural Network potentials (MP-NNPs), outperforming other

machine learning potentials in terms of accuracy [34] [38] [39] [40] [37]. MP-NNPs use a graph neural network [46] to represent the molecular structure, assigning the nodes with atomic properties and the edges with their interactions within a cutoff distance. The latent state of each node i at time t is described by the tuple

$$\sigma_i^{(t)} = (\mathbf{r}_i, z_i, h_i^{(t)}) \quad (2)$$

where $\mathbf{r}_i$ is the three-dimensional position vector of the atom i , z_i is the atomic species, and h_i is a vector storing learnable features. To incorporate into the model that the potential energy of a structure is equivariant with respect to the symmetry group of $E3$ (rotations, translations and inversions), h_i is chosen to be an invariant, two-body tensorial or vectorial property to represent the nodes. This design significantly lowers the computational costs and increases accuracy, as the equivariant features do not have to be learned during the training.

To forward-pass the information in the network, three key steps are being iterated: (a) message construction, (b) update, and (c) readout. The message m_i is constructed for each atom i according to the function

$$m_i^{(t)} = \bigoplus_{j \in \mathcal{N}(i)} M_t(\sigma_i^{(t)}, \sigma_j^{(t)}) \quad (3)$$

where j is the index of the neighbouring atom of atom i , $\mathcal{N}(i)$ is the set of neighbouring atoms, M_t is a learnable message function, and $\bigoplus_{j \in \mathcal{N}(i)}$ is a permutationally invariant, learnable pooling function (usually a sum over the neighbours j). The state of the node is then updated according to the equation:

$$h_i^{(t+1)} = U_t(\sigma_i^{(t)}, m_i^{(t)}), \quad (4)$$

where U_t is a learnable update function. After constructing the messages and updating the nodes for T iterations, the energy of the atoms is constructed with the readout function

$$h_i^{(t+1)} = U_t(\sigma_i^{(t)}, m_i^{(t)}), \quad (5)$$

which sum up to the total energy of the system

$$E = \sum_{i=1}^N E_i, \quad (6)$$

where N is the total number of atoms in the system. [37] By iterating the message-passing steps, the model is able to learn semi-local information beyond the cutoff up to the model’s “receptive field”. The receptive field can be expanded using multiple message-passing

layers, eventually allowing the model to learn non-local information.

2.2. The MACE Architecture

Although message-passing neural network potentials outperform other MLPs in terms of accuracy [34] [38] [39] [40] [37], they have high computational costs when the system is scaled and non-local features are incorporated. [37]. This scaling limitation arises from using two-body node features h_i , which requires using a high number of message-passing layers to incorporate many-body effects. [41] However, the number of neighbouring atoms scales cubically with the number of message-passing layers, making the parallelization challenging and significantly increasing the computational cost. [36] The recently developed Message-Passing Atomic Cluster Expansion (MACE) [37] model overcomes this scalability problem by employing MPNNs with higher body order message constructions:

$$m_i^{(t)} = u_1(\sigma_i^{(t)}; \sigma_j^{(t)}) + \sum_{(j_1, j_2)} u_2(\sigma_i^{(t)}; \sigma_{j_1}^{(t)}, \sigma_{j_2}^{(t)}) + \dots + \sum_{(j_1, \dots, j_v)} u_v(\sigma_i^{(t)}; \sigma_{j_1}^{(t)}, \dots, \sigma_{j_v}^{(t)}), \quad (7)$$

where the functions u are learnable functions, and v is the correlation order hyperparameter such that $v + 1$ gives the body order of the message. Previous models, such as DimeNet [42] [43], have also incorporated higher-order features, however, the self-interactions have been excluded using the summation $\sum_{n, j_1 < \dots < j_v}$. MACE have incorporated the self-interactions as this feature allows for constructing a message tensor product that does not scale exponentially with the correlation order v , unlike DimNet.

This computationally efficient message construction with higher body terms is achieved by first obtaining a two-body feature $A_i^{(t)}$ at each message-passing iteration t of each node i using learnable radial basis functions $R_{kl_1 l_2 l_3}^{(t)}$, learnable spherical harmonics $Y_{l_1}^{m_1}(\hat{\mathbf{r}}_{ji})$, and features from each previous layer $h_{j, \tilde{k} l_2 m_2}^{(t)}$ with learnable weights $W_{\tilde{k} l_2}^{(t)}$:

$$A_{i, kl_3 m_3}^{(t)} = \sum_{l_1 m_1, l_2 m_2} C_{l_1 m_1, l_2 m_2}^{l_3 m_3} \sum_{j \in N(i)} R_{kl_1 l_2 l_3}^{(t)}(r_{ji}) Y_{l_1}^{m_1}(\hat{\mathbf{r}}_{ji}) W_{\tilde{k} l_2}^{(t)} h_{j, \tilde{k} l_2 m_2}^{(t)}. \quad (8)$$

$C_{l_1 m_1, l_2 m_2}^{l_3 m_3}$ are the Clebsch–Gordan coefficients, r_{ji} is the interatomic distance between atoms i and j , and

$N(i)$ is the set of neighbours of atom i . The summation over $N(i)$ ensures permutational invariance, while the Clebsch–Gordan coefficients maintain equivariance. After the construction of $A_i^{(t)}$, the higher body-order tensor product $B_{i, \eta v k L M}^{(t)}$ is obtained by using $A_i^{(t)}$ as the basis with weights $w_{\tilde{k} l_2}^{(t)}$ and symmetrising it:

$$B_{i, \eta v k L M}^{(t)} = \sum_{l_m} C_{\eta v, l_m}^{L M} \prod_{\xi=1}^v \sum_{\tilde{k}} w_{\tilde{k} l_2}^{(t)} A_{i, \tilde{k} l_2 m_\xi}^{(t)}, \quad l_m = (l_1 m_1, \dots, l_v m_v), \quad (9)$$

where $C_{\eta v, l_m}^{L M}$ correspond to the Clebsch–Gordan coefficients, L and M are the total angular and magnetic quantum numbers corresponding to the expansion of $A_i^{(t)}$, and the ηv labels all possible configurations of the $l_1, \dots, l_v$ features. The correlation order v gives the order of the tensor product $B_i^{(t)}$, which can be specified by the user. The message is then expanded using $B_i^{(t)}$ as the basis with learnable weights $W_{z_i}^{(t)}$:

$$m_{i, k L M}^{(t)} = \sum_{v, \eta v} W_{z_i}^{(t)} B_{i, \eta v k L M}^{(t)} \quad (10)$$

where z_i is the atomic species of atom i . As a final step in the MACE architecture, the PES of the system is obtained by mapping the node features $E_i^{(t)}$ at each message-passing step t such that they sum up to the total energy:

$$E = \sum_{i=1}^N \sum_{t=1}^T E_i^{(t)}, \quad (11)$$

where the atomic energies $E_i^{(t)}$ are obtained using the hierarchical readout function

$$E_i^{(t)} = R_t(h_i^{(t)}) = \begin{cases} \sum_{\tilde{k}} W_{\text{readout}, \tilde{k}}^{(t)} h_{i, \tilde{k} 00}^{(t)}, & \text{if } t < T \\ \text{MLP}_{\text{readout}}^{(t)}(h_{i, k 00}^{(t)}), & \text{if } t = T. \end{cases} \quad (12)$$

with invariant node features $h_{i, k 00}^{(t)}$ to ensure that the invariance of atomic energies is conserved.

The key breakthrough in this model is that the tensor product expansion shown in Eq. 9. integrates many-body interactions implicitly, avoiding the calculation of the higher-body sums shown in Eq. 10. This allows the construction of higher-body messages without computational limitations. Since higher-order messages capture more information in a single step, fewer message-passing layers are required to approximate the PES. Additionally, the only non-linear terms in the model are the radial basis functions and the final readout. As a result,

MACE is a fast and highly parallelizable graph tensor network, allowing simulating systems containing 1000 atoms for nanoseconds per day on a GPU, with an accuracy that reaches or exceeds the state-of-the-art benchmarks rMD17, 3BPA, AcAc. [37] [44]

2.3. MACE-MP-0: A Foundation Model for Atomistic Simulations

Machine Learning Potentials have enabled simulations with ab initio accuracy and high computational efficiency. However, state-of-the-art models require significant human and computational resources to develop and validate a potential for a particular system of interest. [15] MACE-MP-0 is a remarkable pre-trained foundational model that aims to overcome this challenge. It uses the MACE architecture outlined in Section 2.2. and is designed to simulate a wide range of atomic systems without the need for system-specific retraining. [44]. This high transferability across different materials and molecular systems is achieved by training the model on the extensive MPtrj dataset, which contains 150,000 inorganic crystal structures. [45]

The remarkable feature of MACE-MP-0 is that it can predict the PES of systems that are not included in the training set with various phases, making it one of the most general MLPs up to date. To name a few examples, MACE-MP-0 has been used to perform stable molecular dynamics simulation of bulk solids, interfaces and amorphous materials, predict phonon spectra, calculate the activation energies of defect migration, or simulate solvent mixtures and reactive processes such as proton transfer in water. The generality of MACE-MP-0 is a transformative step in the development of MLPs, as it significantly lowers the barrier to entry for MLPs in material science. This report attempts to apply MACE-MP-0 to simulate the dynamics of a mini protein, being a good test for the generalizability of MACE-MP-0, as biomolecules have not been included in the training set.

3. Results

3.1. System Details

3.1.1. Chignolin as a Test Case for Protein Dynamics with MACE-MP-0

This report evaluates the capability of MACE-MP-0 to simulate protein dynamics, building upon the protein-dynamics test presented in Section A.7 of the MACE-MP-0 paper [44]. In that study, two mutants of the chignolin protein were examined: the 166-atom 5AWL variant [46] and the 138-atom 1UAO variant [47], both simulated in a vacuum. The structural stability of the proteins were tested by computing the radius of gyration

over the course of the simulation. However, as detailed later in Section 3.1.2, chignolin’s stability is strongly influenced by solvation effects. To address this, the present report extends the original study by performing simulations in both vacuum and solvated environments, allowing a more comprehensive analysis of chignolin’s stability under realistic conditions.

Chignolin is a suitable candidate for testing the ability of MACE-MP-0 to simulate protein dynamics for three reasons. Firstly, it classifies as a mini-protein, falling into the range of the system size that MACE-MP-0 is able to simulate. [48] [44] Secondly, chignolin has a folding-unfolding dynamics in the timescale of 10 ns. Therefore, within the 1ns simulation window that MACE-MP-0 can realistically simulate, a folded initial structure is expected to remain compact, while an unfolded structure should begin to show signs of becoming more compact due to its fast-folding behavior. This makes chignolin a good system for testing MACE-MP-0 to see if it can capture early folding dynamics. Thirdly, chignolin has been used extensively in both in vitro and in silico studies due to its small size, making it a good benchmark for simulations. [49] This report uses the remarkable AIMD-Chig data set for benchmarking, which contains 2 million conformations of the 166-atom size mutant of chignolin, 5AWL, sampled with the ab initio accuracy level of DFT. [48]

3.1.2. The Folding Dynamics of Chignolin

Both 5AWL and 1UAO mutants of chignolin has stable native states folded into a β -hairpin structure that can undergo fast folding-unfolding dynamics. The folding of the β -hairpin is driven by hydrophobic interactions and the pH of the solvent. The β -hairpin is stable due to the hydrogen backbone keeping the β -sheet conformation antiparallel and due to key side-chain interactions, such as Tyr2–Trp9 aromatic interaction and the Tyr2–Pro4 CH– π interaction. A crucial player in stabilising the β -hairpin fold is the protonation of the Tyr2 residue (Figure 1). In vitro research has shown that at a high pH environment, the Tyr2 deprotonates, which disrupts stabilising interactions, leading to a more flexible and disordered structure. [49] Additionally, the folding kinetics of Chignolin are highly sensitive to solvent conditions, as hydration shell layers are formed around the protein, and the water molecules exchange protons with the protein residues and form a hydrogen bonding network, keeping the folding stable. [49]

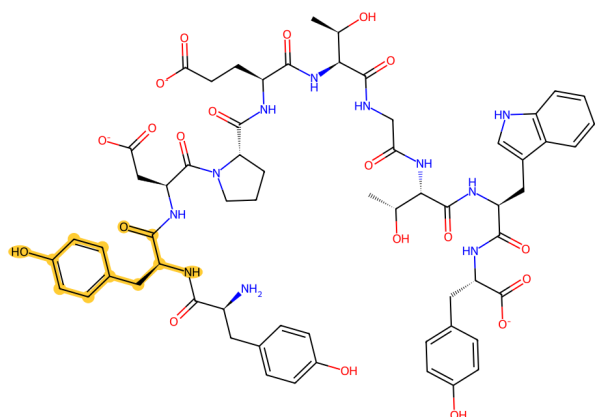


Figure 1: The 166-atom size chignolin protein (5AWL mutant). In yellow, the Tyr2 residue is highlighted, whose protonation is essential to maintain the folded structure. (The figure was created by the author using the Python package RDKit [50])

3.1.3. Simulation Conditions for Chignolin Variants and Observable Calculations

To test the MACE-MP-0, the 166-atom size 5AWL mutant is simulated in both solvated and vacuum environments, using a folded and unfolded initial structure. To assess the protein’s stability, the radius of gyration is calculated over the simulation time. Although the radius of gyration is not a definitive measure to assess if a protein is folded – since deeper structural analysis is required for that, for example, Secondary Structure Analysis [51] – it is a good indicator of how compact the protein is [52], which can be compared to the compactness of the folded 5AWL-AIMD simulation, obtained from the AIMD-Chig dataset. [48]

Additionally to the 5AWL simulations, the 138-atom size 1UAO mutant is also simulated in a vacuum with charged and neutralized residues. This method attempts to simulate the water bulk environment implicitly, as neutral residues “mimic” the solvation effects. In a water bulk, a charged protein would have frequent proton transfers with the first solvation shell of the water surrounding it, which in turn would make it more stable. Therefore, by neutralizing the residues, the protein is put into a state in the PES as if it has already interacted with the water molecules. The 138-atom size mutant was chosen for the simulation of these systems, as due to its smaller size, it equilibrates faster and requires less simulation time for observable conformational changes. Table 1. shows the summary of the system details of the 6 simulations run in total to test different protein dynamics features.

Table 1: Simulation conditions for different chignolin mutants.

Mutant	Structure	Environment
5AWL	folded	solvated
5AWL	folded	vacuum
5AWL	unfolded	solvated
5AWL	unfolded	vacuum
1UAO	folded, neutralized residues	vacuum
1UAO	folded, charged residues	vacuum

3.2. Reproduction Process

3.2.1. Python Environment and Source Code

The following packages have been utilized for the simulation with Python version 3.12.3:

Table 2: The Python packages used in the simulation.

Package	Version
numpy	1.26.4
ase (Atomic Simulation Environment)	3.24.0
mace.calculators (MACE package)	0.3.10
torch (PyTorch)	2.5.1

The MACE architecture was downloaded from: <https://github.com/ACEsuit/mace.git>. To solvate the system, the PackMol package was used that randomly places water molecules around the given input structure. <https://m3g.github.io/packmol/>

3.2.2. Simulation Setup

The input files and code written for the simulation are included in the .tar file named **code.MACE-MP-0 HYP9J.tar**. The INPUTS folder contains the input structures of the mutants 1UAO (neutral and charged residues), 5AWL (folded and unfolded), and a water molecule. The **setup_simulation.py** script initializes the functions for the molecular dynamics simulation.

The first function is **get_simulation_box()**, which calculates the simulation box dimensions by first determining the protein’s bounding coordinates and then adding the specified box margin around the protein (**sim_box_margin**). Then, **write_packmol_script()** generates an input script for the Packmol package that places the protein into the simulation box and randomly distributes a specified number of water molecules (**num_waters**) inside the box with a given packing tolerance. After this step, **solvate_protein()** executes the Packmol input script and saves the solvated protein file structure in .pdb format.

run_NVT_ASE_simulation() executes the ASE molecular dynamics simulation either in a vacuum or

solvated environment (Figure 2). During the simulation, `run_NVT_ASE_function()` records and outputs two files: `trajectory.pdb`, containing the trajectory of the MD simulation, and `log_data.txt`, storing the timesteps, energies, temperature, and the elapsed time of the simulation with a specified sampling frequency (`sampling_freq`). These pieces of information are useful not only for post-analysis but also for monitoring the system’s stability during the simulation.

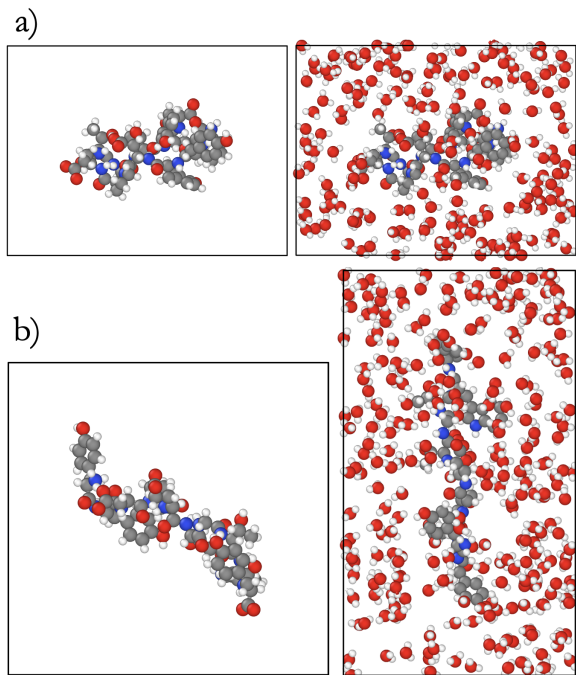


Figure 2: The vacuum (left) and solvated (right) simulation setup of (a) the folded 5AWL mutant, and (b) the unfolded 5AWL mutant.

3.2.3. Parameters and System Specifics

The simulations are run in an NVT ensemble using the ASE MD package at 300K for 1ns with a simulation timestep of 1fs, using Langevin dynamics with a friction coefficient of 0.01 fs^{-1} and periodic boundary conditions. The sampling is done every 1000fs. The structure file of the 1UAO mutant is obtained from the protein data bank [47], and the neutralized version is created by protonating the ASP3, GLU5, GLY10 residues and deprotonating TYR1 using PyMol. [53] The folded and unfolded structure of the 5AWL mutant is obtained from the AIMD-Chig dataset [48].

To run the functions defined in `setup_simulation.py`, the script `run_simulation.py` has to be executed in the terminal. This code prompts the user to specify 8

configurable parameters (shown in Table 3) which are used to set up the simulation.

Table 3: A table showing the parameters of the chignolin simulation, which can be interactively specified by the user when executing the `run_simulation.py` script.

Parameter	Description
<code>protein_name</code>	File name of the protein structure in the INPUT folder (PDB format)
<code>additional_name</code>	Optional name tag to distinguish this simulation
<code>solvated</code>	Boolean: run in solvent (True) or vacuum (False)
<code>sim_box_margin</code>	Extra space (in Å) added around the protein to define the box
<code>num_waters</code>	Number of water molecules to add (only required if <code>solvated = True</code>)
<code>n_steps</code> (=100,000,000)	Number of molecular dynamics steps (1 step = 1 fs)
<code>sampling_freq</code> (=1000)	Interval for saving trajectory and log data (in steps)
<code>output_dir_base</code>	Directory where output files (trajectory, logs) are stored

Table 4: Variables controlling system-specific behaviors and model settings. These are pre-defined in the `run_simulation.py` script, in addition to the user-defined parameters shown in Table 3.

Variable name	Description
<code>tolerance = 2</code>	Packmol packing tolerance for solvation setup, minimum allowed distance between molecules (Å)
<code>model = 'medium'</code>	MACE potential model used for the simulation
<code>dispersion = False</code>	Enables dispersion interactions in MACE if set to True
<code>enable_cueq = False</code>	Enables charge equilibration in MACE if set to True
<code>T_init = 300</code>	Initial temperature of the simulation (K)
<code>timestep = 1.0 * units.fs</code>	Simulation timestep (fs)
<code>friction = 0.01 / units.fs</code>	Friction coefficient for Langevin dynamics (fs^{-1})

In addition to the user-defined parameters shown in Table 3, the `run_simulation.py` script contains pre-defined variables, listed in Table 4, that further specify simulation parameters.

The number of water molecules and the simulation

box margin used in the simulations are summarized in Table 5. These numbers have been optimised manually with a trial-and-error methodology so that the water molecules form a coherent solvation shell around the protein and the system size is not exceeding the computational limit of the MACE architecture.

Table 5: The parameters used to set up the solvated environment. The simulation box margin is the extra space (in Å) added around the protein for defining the simulation box.

Mutant	Structure	Number of Water Molecules	Box Margin (Å)
5AWL	folded	200	7
5AWL	folded	0	7
5AWL	unfolded	250	8
1UAO	standard	0	8
1UAO	neutralized	0	8

3.2.4. Processing the Simulation

The script `process_simulations.py` calculates the gyration of radius of the protein using the function [52]

$$R_g = \sqrt{\frac{\sum_i m_i (\mathbf{r}_i - \mathbf{r}_{\text{COM}})^2}{\sum_i m_i}}$$

and saves the radius of gyration values as a `.txt` file. The water molecules and detached atoms from the residues are excluded from the radius of gyration calculation (see more details in Section 3.3). The indices of the detached atoms were extracted manually by analysing the trajectories. The **Process_AIMD.py** function calculates and plots the radius of gyration of the AIMD-Chig reference data [48], and the **process_rg_comparison.py** script plots the radius of gyration versus simulation time for all simulations in a single figure.

3.3. Results and Observations

3.3.1. 5AWL Folded Simulations

Figure 3. presents the results of the calculations of the radius of gyration of the 5AWL and 1UAO variants at different conditions. The 5AWL-AIMD reference data shows a stable R_g value of approximately 6.5 Å with minor fluctuations, which is characteristic of a well-folded β -hairpin [49]. The 5AWL-MACE-MP-0 folded simulations captured this trend closely, with an average deviation of less than 0.2 Å, suggesting that the model maintains structural integrity similar to AIMD. This highlights the remarkability of MACE-MP-0 in its ability to simulate protein dynamics.

On the other hand, the solvated, folded 5AWL-MACE-MP-0 shows a trend of becoming less compact, which contradicts the expectation that solvation effects should stabilise the chignolin structure (Section 3.1.2). However, it is important to note that this simulation only ran until 600 ps due to the CUDA environment running out of memory. Therefore, the observed upward trend in the gyration radius may be just a transient fluctuation rather than an unfolding event.

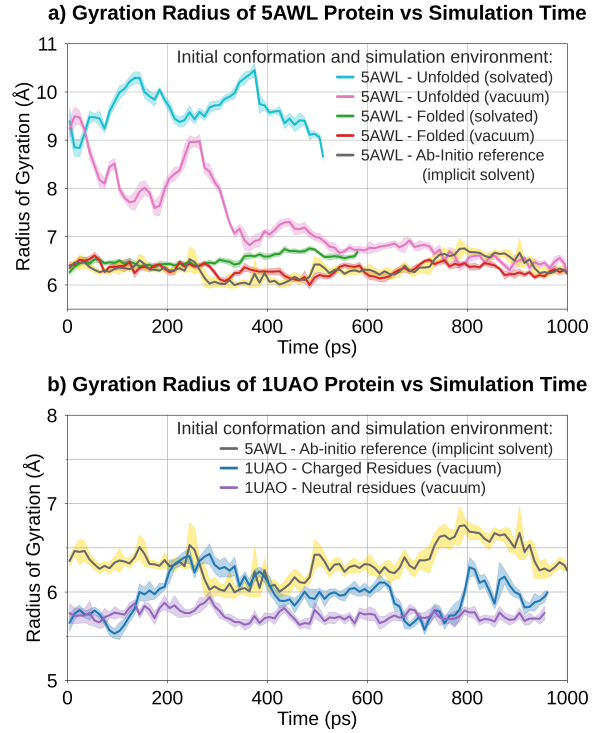


Figure 3: A figure showing the change in compactness of the chignolin protein mutants 5AWL and 1UAO with different initial conformations over the simulation time, using the MACE-MP-0 potential [44]. The simulations for the 5AWL variants have been performed in vacuum and solvated environments. The compactness of the proteins is characterised by the radius of gyration [52], and compared to the 5AWL-AIMD reference dataset [48]. The shaded areas represent the standard deviation of values averaged over 10 ps.

3.3.2. 5AWL Unfolded Simulations

Analysing the dynamics of the unfolded 5AWL variant in a vacuum, it can be observed that the system converges towards its more stable folded native structure, further demonstrating MACE-MP-0 ability to simulate protein dynamics. However, similarly to the folded-solvated 5AWL experiment, the solvated system does not follow this trend. Since the simulation only ran until 500 ps, the conclusion remains the same as in Section 3.3.1: this behaviour may reflect a transient fluctuation.

3.3.3. 1UAO Simulations

It can be seen in Figure 3. that the simulations of the 1UAO mutant with neutralized and non-neutralized residues successfully preserved the compactness of the native structure. However, comparing the neutralized and non-neutralized unsolvated simulations, it can be observed that the neutralized mutant demonstrated a greater stability, supporting the argument that by neutralizing the residues the system is put into a more physical part of the PES (see Section 3.1.1).

3.3.4. Analysing the Behaviour of Solvated Systems

There are four possibilities for the observed upward trends in the solvated simulations: a) it is only a transient fluctuation, b) it is a result of MACE-MP-0's limitation, c) the system has not equilibrated under the simulation time or d) the solvated system is unphysical. Analysing the potential energy of the system (Figure 4), it can be seen that equilibration occurs around 250 ps; therefore, possibility c) can be ruled out. Examining the end-trajectories of the simulations reveals that the solvation is not executed perfectly, as water droplets form instead of a bulk system (Figure 5). Therefore, it is strongly supported that the solvated simulations did not behave as expected due to reason d) – the solvation methodology outlined in Section 3.2.2 did not perform well. Either more water molecules are needed into the system, or a smaller simulation box with a higher density of water molecules.

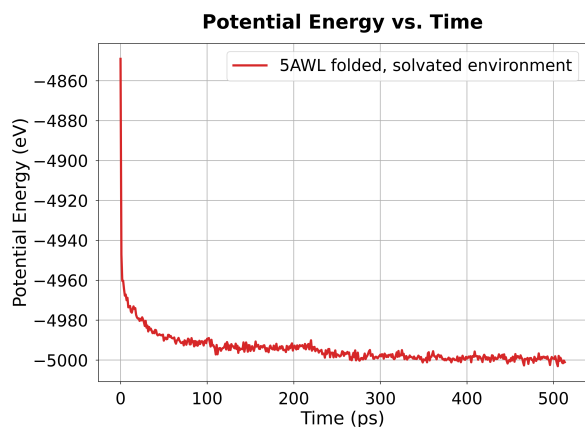


Figure 4: A figure showing the change in the potential energy of the 5AWL chignolin simulations in solvated environments.

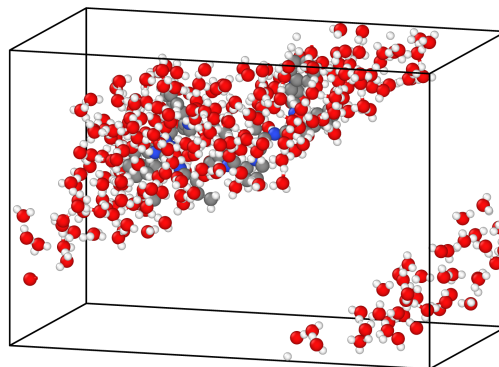


Figure 5: A snapshot of the last frame of the 5AWL-unfolded simulation in a solvated environment, depicting the simulation cell with periodic boundary conditions.

3.3.5. Protonation Effects

An important result of the simulations is that protonation interactions are observed when examining trajectory data (Figure 6). This finding demonstrates the remarkable feature of MACE-MP-0 that it can simulate reactive processes with system sizes with approx. 1000 atoms.

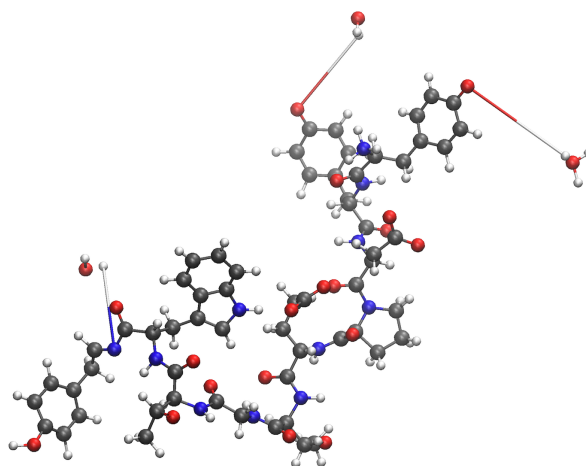


Figure 6: A snapshot from the simulation of the 5AWL folded mutant in a solvated environment, capturing a protonation effect.

3.3.6. Detached Molecules

Further examination of the trajectory data reveals some unphysical behaviour. In the folded-5AWL simulations, the atoms with indices 80, 81, 82; in the unfolded-vacuum-5AWL simulations, atoms with indices 145, 144, 146; and in the unfolded-solvated-5AWL simulations, atoms with indices 163, 165, 164 are observed to be detached from the protein. These atoms consistently correspond to one carbon and two

oxygen atoms. An example of this can be seen in Figure 7. This behaviour appears to be independent of the environment and initial structure, suggesting that it may be a limitation of the MACE-MP-0 potential. However, no detachments have been observed in the case of 1UAO simulations. Therefore, additional test simulations would be necessary to confirm this conclusion.

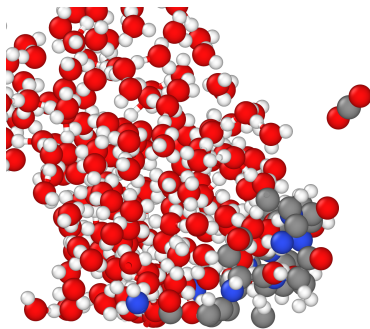


Figure 7: A snapshot of the 5AWL chignolin mutant’s simulation in a solvated environment, showcasing that a CO₂ molecule has separated.

3.4. Comparison of the Results with MACE-MP-0 Paper and Experimental Studies

Figure 8. depicts the protein-dynamics findings from the MACE-MP-0 paper [44]. Comparing the original MACE-MP-0 study with the simulations presented in this report (Figure 3), it can be established that the key trends are successfully reproduced. Both Figure 3. and Figure 8. demonstrate the stability of the folded structures and increasing compactness of the unfolded structures in a vacuum, effectively showcasing the success of MACE-MP-0.

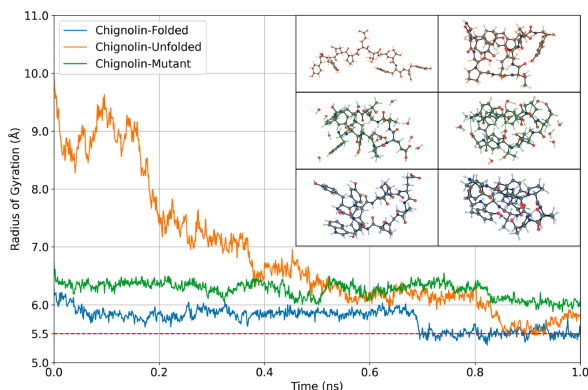


Figure 8: Results from the MACE-MP-0 paper. Chignolin-Mutant refers to the structure of the 5AWL mutant plus 12 crystallographic water molecules, obtained from the protein data bank. Chignolin-Folded / Unfolded refers to the 1UAO chignolin mutant, obtained from the AIMD reference dataset [44].

4. Conclusions

4.1. Strength of the Approach Used

This work explored the ability of the pre-trained, foundational potential MACE-MP-0 [[44] to simulate complex protein dynamics of the chignolin mini-protein, which constitutes a substantial test to evaluate the model’s extrapolation capabilities and its generalization, as biomolecules were not included in the model’s training set. The study has successfully demonstrated MACE-MP-0’s capability to produce a stable simulation of the native structure of chignolin and, remarkably, to capture its early folding dynamics. The main strength of the model is that it was able to simulate proton transfers between the protein residues and the first solvation shell of the water surrounding it. This showcases the strength of MACE-MP-0 as a reactive MLP capable of capturing intricate biomolecular interactions, as this achievement remains challenging for conventional non-reactive force fields or ab initio simulations due to system size limitations.

4.2. Limitations

The main limiting factor of the MACE-MP-0 is the maximum system size it can simulate. Currently, the MACE architecture’s computational capacity is a system size with a maximum of 1000 atoms. However, this report has reached this plateau, as the solvated-5AWL system consisted of 916 atoms in total. This computational limitation was also visible as the solvated simulations did not run until 1 ns opposed to the vacuum simulations, due to the CUDA running out of GPU memory. This computational limitation affected the accuracy of the simulations, as only water droplets could be set up with the available number of water molecules instead of a continuous bulk solvent, which can introduce artefacts in the simulation. Additionally, a correct bulk environment is essential to capture solvation effects of the protein dynamics properly, such as proton exchange with the solvation shells, hydrogen bond formation and the pH of the solvent.

A further limitation of the MACE-MP-0 applicability to the proteins is the unphysical behaviour of atoms detaching from the protein. To decipher the features of the MACE-MP-0 responsible for this unphysical behaviour, further tests are needed to investigate the underlying cause of the malfunction. For example, as a first step it would be beneficial to test the same simulation setup for the 5AWL protein with a traditional force-field to assess if the problem is with the MACE-MP-0 potential or with the simulation set up.

4.3. Future Outlook

Several modifications could be made to improve the accuracy of the simulations. Firstly, ions can be introduced to the system to define a non-neutral pH environment. This modification would create a more physical environment for chignolin, capturing more accurate protein-folding dynamics. Additionally, the stability of the B-hairpin could be observed at different pH conditions, providing further atomistic insights into protein folding dynamics. Secondly, it would be intriguing to remove the proton from the Tyr2 residue (shown by in vitro experiments that play a crucial role in the stability of the -hairpin structure) and confirm whether the structure breaks as physically expected.

Another further improvement to the simulation setup is refining the solvation methodology. The current approach defines the number of water molecules and the size of the simulation box with a manual trial-and-error method, where the system is allowed to condense independently. However, techniques exist that create a solvation environment more efficiently and ensure that a bulk-water system is created instead of water droplets. For example, one technique is to shrink the simulation box using its wall as pistons to condense the protein-water system until it reaches the target density. Another approach adds dummy atoms around the protein-water system and assigns the atoms with forces that point towards the centre of the simulation box. Then, these dummy atoms act as pistons and compress the protein-water system. When the target density is reached, the dummy atoms are removed, and the system is equilibrated. A third approach for solvation would be to apply a pulling force to the protein-water system to compress itself. However, this methodology could put the protein into a metastable conformation, prolonging the equilibration period, decreasing computational efficiency. A further improvement of this study is to fine-tune the MACE model to the chignolin's structure or employ a machine learning potential trained on a data set containing biomolecules.

In conclusion, this study demonstrates that MACE-MP-0 can simulate a mini-protein's structural stability and early folding dynamics and capture crucial, reactive processes such as protonation effects. The key limitation of the approach is that the MACE-MP-0 potential cannot simulate system sizes larger than 1000 atoms, which makes it challenging to create a bulk-water-solvated environment with consistent bulk conditions. However, if this limitation is addressed and improved in the future work of developing foundational models using the MACE architecture, it will become a

powerful tool for simulating reactive biomolecular systems, eventually lowering the barrier of entry for using molecular dynamics simulations in biological experiments due to the foundational model's remarkable generalizability.

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